Lesson notes · for you

Lists and loops

Programming · Level 3 · Lesson 2

Time
20 min

Materials
A phone with this lesson downloaded; A notebook for a week of readings

Learning goals

- Index a list from 0 and from the end with -1.
- Count, total and find the largest in a loop.
- Use a list as a stack or a queue.

Activity: The human list

1. Warm-up (3 min): Five classmates stand in a row holding temperatures, including a negative one. Label them 0 to 4.
2. Index (7 min): Call out temps[0], temps[-1] and temps[1:3]. The right classmates step forward. What does temps[5] do?
3. Loop (7 min): One person is best and holds the first card. Walk along the row; swap whenever you meet a bigger number. Why not start at 0?
4. Stack (3 min): Pile three books and take them off one at a time: the last on is the first off. That is a stack.

Use made-up readings, and include a negative one: it is the case that breaks “start at 0”.

Discussion

- Why does largest start with the first item, not 0?
- Where do you meet stacks and queues in real life?
- What happens when you ask for an index that doesn’t exist?

Check yourself

- Write a loop that counts the even numbers in a list.

Try it this week
Write down five numbers from your day (minutes on the bus, glasses of water…) as a Python list, and find the largest without max().

Worksheet

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Name _____ Class _____ Date _____

1 **hot counts the readings above 30. What is hot at the end?**

Use the code from question 3, with its gaps filled in.

◯ 2 ◯ 4 ◯ 3

2 **Write largest(values): it returns the biggest value in the list, without using max(). It must work for negative numbers too.**

```
def largest(values):
    # biggest value, no max()
    best = 0
    return best
```

Write your program:

3 **Fill the gaps so the program counts the readings above 30.**

```
temps = [21, 31, 35, 30]
hot = 0
for t in _________:
    if t _________ 30:
        hot = hot + 1
print(hot)
```

Word bank: `range` `>=` `temps` `>`

4 **This loop should visit every position in temps, but it stops with an IndexError. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  temps = [21, 25, 31, 28]
2  for i in range(1, len(temps) + 1):
3      print(temps[i])
```

5 **An undo history is a stack. You type A, then B, then C, and press Undo twice. What is left?**

◯ A ◯ B and C ◯ C

✎ **Which data do you collect, or could collect, as a list (steps, scores, temperatures)? Write three questions a loop could answer about it.**

Check your answers

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1 **2**

31 and 35 are above 30; 30 itself is not above 30.

2

```
def largest(values):
    best = values[0]
    for v in values:
        if v > best:
            best = v
    return best
```

Keep the biggest value seen so far and replace it when a bigger one comes.

3

```
temps = [21, 31, 35, 30]
hot = 0
for t in temps:
    if t > 30:
        hot = hot + 1
print(hot)
```

for t in temps visits every reading; > counts only those above 30.

4 **Line 2: `for i in range(len(temps)):`**

Python reports the error on line 3, but its cause is line 2.

5 **A**

Last in, first out: the first Undo removes C, the second removes B, and A is left.